

“Since using Equator we have not made a single bad part and have reduced part production cost by 27%”

The quality of the products a company produces, and their assurance of that quality, can mean all the difference to successfully acquiring contracts. Maintaining that quality also significantly increases the chances of repeat business. High-Tech Engineering, a precision engineering company based in Dunstable, Bedfordshire, UK, has always focused on the quality of the parts it produces. Now with its latest addition, a Renishaw Equator gauging system, it is reaching 100% part inspection and zero scrap whilst halving the operator requirements and reducing part production costs by 27%.

Started in 1985 by Managing Director, Steve Tickner, High-Tech Engineering built a reputation in the motorsport industry for delivering high quality machined parts. The company has since moved into the aerospace sector and gained some key industry approvals, including becoming a preferred supplier to Rolls-Royce and BAE Systems.

Recently High-Tech won a contract to produce precision milled titanium parts for a large aerospace customer. Due to the nature of the parts, High-Tech was instructed to carry out 100% part inspection. Steve Tickner explains, “We knew from the start that we would need to find an inspection method which could not only meet the cycle time requirements for the part, but would be a cost effective solution for us as well. Relying on the CMM we already had wasn’t going to be an option. We couldn’t risk any bottlenecks. We knew that we would either need another CMM or something else which could give us the measuring capacity. That’s what led us to the Renishaw Equator.”

Equator™ gauging system acquisition

It was through a simple internet search that Steve came to learn of the Equator gauging system. After some research and following meetings with representatives from Renishaw, High-Tech was impressed with what the Equator gauging system could offer, and was particularly pleased with the overall price of the system.

Commenting on why High-Tech bought the Renishaw Equator gauging system, Steve says: “Space on our factory floor is at a premium. Within the space we set aside for this cell we had to include faster, more efficient technology. This meant that a co-ordinate measuring machine (CMM) with a temperature controlled environment was far too big to be practical. The Equator suited the space perfectly as it’s a compact machine. The added bonus is the fact it is thermally insensitive and doesn’t require any air supply, meaning we didn’t need to spend additional time, money and effort putting in another temperature controlled room or extra piping.”

High-Tech is currently using the Renishaw Equator gauging system to perform 100% inspection of a complex aerospace part. It works by comparing the manufactured parts against a matching master part, gauging all the features in a single operation with an immediate pass/fail decision, along with a report of the component dimensions.

Comprehensive measurement in half the time of the existing CMM option

The Equator gauging system is used to gauge around 150 features including a number of bores, thicknesses and form measurements on the part with typical tolerances of $\pm 25 \mu\text{m}$. The Equator gauging system does this within 10 minutes and well within the production requirements, far less than the machining time. This is almost a 50% reduction in cycle time compared to running the program on High-Tech's CMMs.

Process control, run by the machine tool operator

Production at High-Tech runs 24 hours per day, 5 days per week and is split into 2 overlapping shifts. During this time Equator is used by a number of operators with varying skill sets. Previously, High-Tech Engineering would have relied on its CMMs, housed away from the machine tool in a temperature-controlled quality room. Using the CMM for the latest job would have involved two separate operators, firstly the operator machining the part and secondly the operator with the specialist skills to operate the CMM. By using an Equator instead, newly trained staff can use dimensions from a certified component, approved by the QC operators, to "zero" the Equator gauging system and set its accuracy for subsequent measurements. This has allowed the same operator who machined the part to measure the part, halving the operator time required.

Flexibility and future proofing

The Equator™ gauging system is fully programmable and can be used on multiple parts, meaning High-Tech Engineering can perform highly repeatable and rapid automated routines across numerous contracts resulting in significantly reduced labour costs.

Ease of use – push-button simplicity

Commenting on this, Mr Tickner states, "All of the operators can use Equator. It really is easy; just load the part and push the button. It completely simplifies the process and frees up manpower, which in turn helps to deliver ROI very rapidly for the cell."

[High-Tech has managed to reduce the cost of producing the aerospace part by 27% with the Equator gauging system being a factor in that. This has had a real impact on the competitiveness of this sort of production, allowing the company to make the same precision quality parts, whilst also delivering better value to its customers. It is through implementing these types of solutions and continually striving to improve their processes that High-Tech has been awarded 6 consecutive SC21 silver awards, which is unrivalled in the UK. This award recognises excellence within the aerospace and defence industry, for the quality of the products a company produces, as well as its ability to supply them. To receive a silver award a company must consistently achieve at least 95% for delivery performance and at least 99.5% on quality of parts to all of its customers.](#)

[Mr Tickner explains “Since we started using Equator we have not made a single bad part. The Equator is allowing the operators to operate complete process control. They all look at the Process Monitor screen, part of the Equator software, which lists all of the features being gauged. Next to the feature name there is a little bar which turns from green to amber to red as the size or location for each feature begins to drift. The operators know which tool on the machine is responsible for each feature and so can take corrective measures to occasionally tweak the process, to come back well within tolerance. As we carry out 100% inspection, it would be very hard to make a part wrong.” He continues, “We are also using the gauge data to trial different types of tool, to see which gives us the best level of efficiency – something which might save us more money in the future.”](#)

[Outlook](#)

[Following the success of this manufacturing cell, Mr Tickner plans to use the Equator gauging system as part of future cells he will be commissioning for jobs in the pipeline. He concludes: “When you find something which helps you make a perfect part every time, reduces manpower commitments, reduces overall costs and doesn’t cost a fortune itself, it’s a winning solution. All of the benefits are passed on to our customers who know they are getting the best quality at the best price”.](#)

www.renishaw.com/gauging

www.high-tech-eng.co.uk

www.adsgroup.org.uk

[Caption text](#)

[Two parts produced by High-Tech engineering](#)

[Renishaw Equator gauging system at High-Tech engineering](#)

[The quality control room at High-Tech engineering](#)

[The factory floor at High-Tech engineering](#)

Science at Work! Primary Tour

Tour Outline



Your tour

The Science at Work! Primary Tour is conducted at ANSTO's Lucas Heights campus in Southern Sydney. It is offered to school groups in Years 3-6 (NSW Stages 2 and 3) and takes 2.5 hours, or 3 hours for large groups.

This tour experience assumes no background science knowledge. Students will be given opportunities to ask questions to clarify their understanding or share any thoughts and knowledge. Our staff are passionate and experienced science-qualified education specialists. They aim to engage primary students (and their teachers!) to inspire a life-long interest in science and learning.

For further enquiries or to book a tour for your primary group, contact the ANSTO Education Team:

Phone: 02 9717 3090 — email: tours@ansto.gov.au

Science at Work! Primary School Tour Outline

The primary school tour experience starts at the ANSTO Discovery Centre at Lucas Heights. This large room, open to the public, contains interactive displays and science information and the new ANSTO Virtual Reality (VR) experience.

Students are given some time to engage with the displays and VR headsets upon arrival, before taking a seat in the Discovery Centre Theatre.

Inside the theatre the major functions of ANSTO, as a large public research organisation, are introduced and the important role of science in society is discussed with students. The world of atoms is introduced and some of the solids, liquids and gases making up the Periodic Table of Elements discussed. Students learn how to “build atoms” in an engaging and interactive lesson.

With lights turned off, an enjoyable and informative selection of Colour and Light demonstrations are conducted with student participation. Light is introduced as a small visible portion of the energy rays that surround us. Properties of light, including reflection, refraction, colour mixing, and glowing materials, are explained.

After a 15-minute break (longer if requested), students commence the site tour. Travel is by bus around ANSTO's 70-hectare site, visiting 2-3 key science facilities.

A live camera is used to view the OPAL reactor and students learn about the importance of nuclear medicine in modern society. Students also visit a key research facility that uses neutron beams produced from the reactor to look inside materials and structures. If time permits, students may also visit one of our four large particle accelerator machines, which are used for carbon-dating material that was once living and for important environmental applications.

SCHEDULE (GUIDE ONLY):

30 mins	Introduction to ANSTO & interactive science lesson: “A World of Atoms”
30 mins	Colour and Light Show
15 mins	Break
75 mins	Site tour, visiting 2-3 key science facilities

Contact the Discovery Centre by phone (02 9717 3090) or email (tours@ansto.gov.au) to discuss how we can accommodate the needs of your group.

~~Links to the NSW Science and Technology K-6 Syllabus (2017)~~

~~The Science at Work! tour gives students an overview of how a large scientific organisation operates across several areas of science and technology and includes different occupations. The content provided is of a broad nature and links to several areas across the science syllabus for Stages 2 and 3 (see below).~~

~~Stage 2 – Living World~~

~~Classification of living things~~

- ~~• identify that science involves making predictions and describing patterns and relationships (ACSHE050, ACSHE061) SciT~~

~~Survival of living things~~

- ~~• describe how living things depend on each other and the environment to survive (ACSSU073) SysT~~

~~Stage 2 – Material World~~

~~Changes of state~~

- ~~• identify solids, liquids and gases as states of matter SciT~~
- ~~• recognise that a change of state can be caused by adding or removing heat (ACSSU046)~~

~~Materials are used for a specific purpose~~

- ~~• investigate how the properties of natural and processed materials influence their suitability and use in products, services and/or environments, for example: (ACSSU074, ACTDEK013) DesT SciT~~
 - ~~=elasticity~~
 - ~~=thermal conductivity~~
- ~~• identify the roles of people working in science and technology occupations (ACTDEK010)~~

~~Stage 2 – Physical World~~

~~Energy makes things happen (heat, light and electricity)~~

- ~~• investigate the behaviour of light, for example: (ACSSU080) SciT~~
 - ~~=light reflecting in a mirror and on a variety of different surfaces~~

~~=shadows resulting from interruption of light by an object~~

- ~~• explore some common sources and uses of electrical energy and describe different ways electrical energy can be generated sustainably (ACSSU219)~~

Stage 3 – Living World

~~Growth and survival of living things~~

- ~~• understand that scientific and technological knowledge is used to solve problems and inform personal and community decisions (ACSHE083, ACSHE100) SciT~~

Stage 3 – Material World

~~States of matter~~

- ~~• investigate and compare the properties of solids, liquids and gases (ACSSU077) SciT~~

~~Properties of materials determine their use~~

- ~~• investigate characteristics and properties of a range of materials and evaluate the impact of their use (ACTDEK023) DesT SciT~~
- ~~• identify and evaluate the functional and structural properties of materials (ACTDEK023)~~

Stage 3 – Physical World

~~Transfer and transformation of energy~~

- ~~• identify different types of energy transformations, for example: (ACSSU097)
=heat energy to light energy~~
- ~~• investigate how electrical energy can be transferred and transformed in electrical circuits and can be generated from a range of sources (ACSSU097) SciT SysT~~

~~Forces and energy in products and systems~~

- ~~• describe examples where light, sound, heat and electrical energy transform from one type of energy to another ComT SysT~~